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SMC-LUD: Large-Scale B-Mode Liver Ultrasound Dataset for Hepatocellular Carcinoma and Hemangioma Classification

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Hepatocellular carcinoma (HCC) is a leading cause of cancer-related mortality globally, and accurate classification of liver lesions using ultrasound remains challenging. We present SMC-LUD (Samsung Medical Center - Liver Ultrasound Dataset), a publicly available dataset of B-mode liver ultrasound images collected from Samsung Medical Center, Seoul, Korea, between 2015 and 2024. The dataset comprises 5,385 anonymized ultrasound images from 1,021 patients, categorized into two clinically relevant classes: hepatocellular carcinoma (images = 2,716) and hemangioma (images = 2,669). All HCC cases were histopathologically confirmed through surgical resection or biopsy, while hemangioma cases were radiologically diagnosed based on characteristic imaging features. Each image was labeled and verified by board-certified radiologists and pathologists. The dataset is organized with patient-level grouping. This resource addresses the scarcity of large, well-annotated ultrasound datasets for liver lesion classification and provides a valuable foundation for developing and validating deep learning models in liver cancer screening and diagnosis.

Background and Summary

Hepatocellular carcinoma (HCC) is the most common primary liver cancer and remains one of the leading causes of cancer-related mortality globally¹. Early detection and accurate classification of liver lesions are critical for effective clinical decision-making and improving survival outcomes. B-mode ultrasound is widely adopted as a first-line screening tool for HCC and other liver diseases due to its non-invasiveness, affordability, and ease of use in routine surveillance of high-risk populations². It offers advantages over other imaging modalities like computed tomography (CT) and magnetic resonance imaging (MRI), such as minimal infrastructure and lower costs. However, distinguishing between benign liver hemangiomas and malignant HCC lesions using ultrasound alone remains a significant diagnostic challenge, particularly with small lesions or heterogeneous liver parenchyma³. Consequently, experienced radiologists often rely on adjunct imaging, such as contrast-enhanced CT or MRI for definitive classification⁴.

In recent years, deep learning-based computer-aided diagnosis (CAD) systems have emerged as promising tools for augment radiologists' performance in liver lesion classification, with convolutional neural networks

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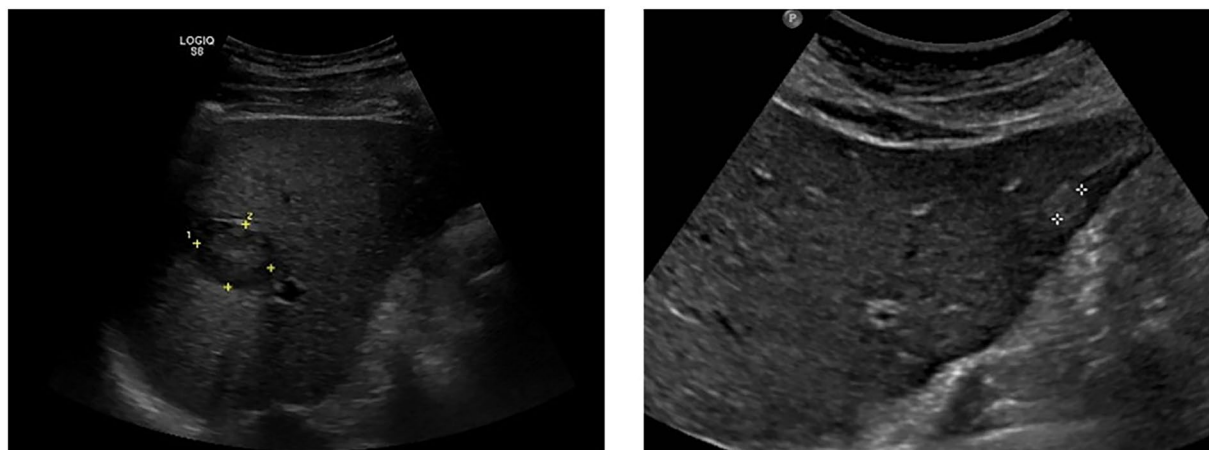


Fig. 1 Representative B-mode ultrasound images in SMC-LUD. (a) Histopathologically confirmed hepatocellular carcinoma (Left). (b) Radiologically diagnosed hemangioma (Right).

(CNNs) demonstrating expert-level sensitivity and specificity in multiple studies⁵. Despite this progress, the development of robust and generalizable deep learning models is often hindered by the scarcity of large, diverse, and well-annotated public ultrasound datasets comprising only a few hundred to a few thousand images, often from single institutions or imaging devices^{6,7}. Furthermore, most open datasets are limited in size and diversity, may focus on narrow clinical scenarios or healthy volunteers, and frequently lack a comprehensive range of lesion types or sufficient ground truth confirmation, especially for HCC cases.

Several publicly available ultrasound datasets for liver lesion analysis exist, but vary considerably in scope and quality. The Cancer Imaging Archive (TCIA) B-mode-and-CEUS-Liver dataset contains approximately 120 patients with various liver masses⁸, while the SYSU-CEUS dataset includes 353 focal liver lesions (186 HCC, 109 hemangioma, 58 focal nodular hyperplasia)⁹. These datasets have contributed valuable resources to the field, yet they remain limited in scale or focus primarily on contrast-enhanced ultrasound (CEUS) rather than B-mode imaging. Our dataset addresses these gaps by providing 5,385 B-mode images from 1,021 patients, all with robust ground truth: histopathological confirmation for HCC cases and definitive radiological diagnosis for hemangiomas. The binary classification task between HCC and hemangioma, while seemingly straightforward, represents a clinically critical distinction, as ultrasound alone often requires adjunct imaging modalities for confident differentiation. Recent studies employing deep learning on similar binary liver lesion classification tasks report accuracies ranging from 83% to 93%^{3,10}, underscoring the diagnostic challenge of ultrasound-based differentiation.

To address these limitations, we present a newly curated and publicly available dataset, SMC-LUD, which consists of B-mode liver ultrasound images collected from real-world clinical settings at Samsung Medical Center (SMC), Seoul, Korea, between 2015 and 2024. The dataset comprises 5,385 anonymized ultrasound images categorized into two clinically relevant classes: HCC (images = 2,716) and hemangioma (images = 2,669). All HCC cases were histopathologically confirmed through surgical resection, while hemangioma cases were radiologically diagnosed based on characteristic imaging features. All data were fully anonymized in accordance with ethical standards and institutional review board (IRB) protocols.

This resource is intended to address the scarcity of large, well-annotated ultrasound datasets for liver lesion classification. It provides a valuable foundation for developing and validating deep learning models aimed at improving liver cancer screening and diagnosis. Furthermore, this work introduces this dataset to the research community and briefly presents the development of an attention-enhanced CNN framework as a baseline model to demonstrate its utility in ultrasound-based liver lesion classification for clinical decision support in liver cancer care.

Methods

Data acquisition and preprocessing. The dataset comprises B-mode ultrasound images retrospectively collected from patients who underwent abdominal ultrasound examinations for liver screening at Samsung Medical Center (SMC) between 2015 and 2024. Ethical approval for this study, including the use of anonymized patient data, was obtained from the SMC IRB (File No. 2024-10-115). Inclusion criteria were as follows: (1) for the HCC group, patients with histopathologically confirmed HCC through surgical resection or biopsy; (2) for the hemangioma group, patients with radiologically diagnosed hemangioma based on characteristic imaging features and explicit documentation by experienced radiologists in the radiology report, without the need for histopathological confirmation.

For HCC cases, a single representative 2D B-mode ultrasound image demonstrating typical HCC characteristics was selected from the period proximal to the date of surgical resection or biopsy. For hemangioma cases, representative images were selected based on definitive radiology reports and corresponding imaging findings. All ultrasound images selected for the dataset were subsequently reviewed and confirmed by at least two board-certified abdominal radiologists to ensure appropriate representation of the respective diagnosis category.

At a tertiary hospital, abdominal ultrasound is routinely performed as a screening tool for liver disease, including early detection of liver cancer [Fig. 1]. The approach to case confirmation distinguishes between

hepatocellular carcinoma (HCC), which often requires histological confirmation, and hemangioma, which is typically diagnosed non-invasively through imaging without surgery or biopsy.

To prepare the data for model training and validation, all images were resized to 224×224 pixels and normalized to zero mean and unit variance based on the training dataset statistics. Data augmentation techniques were applied to the training set to enhance model generalization, including horizontal and vertical flipping, slight rotation, and minor adjustments in brightness and contrast. Although both Clean and Caliper versions of the dataset are publicly available, only the Clean version was used for model training and validation to avoid potential learning of spurious visual cues introduced by caliper overlays. The dataset was split at the patient level into train (70%), validation (20%) and test (10%) sets, ensuring that images from the same patient did not appear in both splits, thereby preventing data leakage.

Model architecture. The proposed model builds on the ResNet-50 backbone¹¹, which was modified to include a parallel attention structure [Fig. 2]. ResNet-50 was selected as the foundational architecture due to its proven effectiveness in medical image analysis and residual learning framework that facilitates training of deep networks.

After feature extraction by ResNet-50, the network splits into two parallel channel-attention pathways designed to minimize information loss.

A lightweight 3×3 convolution first reduces the backbone output from 2048 to 1024 channels while preserving spatial resolution. From this shared tensor, two branches run concurrently with cross-ordered channel attention.

Branch-1 (SE $\rightarrow$ Channel Attention): The feature map is first processed by a squeeze-and-excitation (SE) block⁴ to perform global channel recalibration, and then reweighted by a channel attention module. This ordering allows globally calibrated channel responses from SE to be further refined through channel attention, enhancing discriminative feature selection.

Branch-2 (Channel Attention $\rightarrow$ SE): The same input is first reweighted by a channel attention module and subsequently passed through an SE block. This order highlights discriminative channels prior to the SE's squeeze-and-excite gating, capturing dependencies that benefit from early channel emphasis.

By concatenating the two 1024-channel outputs along the channel dimension, the model preserves complementary cues that either ordering might suppress, thereby reducing information loss.

After the merge, a spatial attention module is applied to the 2048-channel tensor to compute a spatial attention map and gate features element-wise, refining location-specific responses^{12,13}. Finally, global average pooling aggregates the attended features, and a fully connected layer maps the 2048-dimensional vector (concatenated 1024×2) to the target classes.

Training procedure. The model was trained using the Adam optimizer with an initial learning rate of 3×10^{-5} and a batch size of 32 for 100 epochs. Early stopping based on validation loss was applied. To prevent shortcut learning driven by spurious correlations, the model was trained on a clean version of the dataset excluding images containing calipers, rather than on the full dataset.

Baseline models including DenseNet-121¹⁴, VGG-16¹⁵, ResNet-18 and ResNet-50¹¹, EfficientNet-B0¹⁶, and Swin Transformer (Tiny)¹⁷ were trained under the same experimental settings. All models were implemented in PyTorch and trained on NVIDIA 4060 GPUs.

Data Records

SMC-LUD¹⁸ is a dataset of B-mode liver ultrasound images retrospectively collected at Samsung Medical Center (SMC), Seoul, Korea, between 2015 and 2024. The dataset comprises 5,385 anonymized 2D ultrasound images categorized into two clinically relevant classes: hepatocellular carcinoma (HCC; images = 2,716) and hemangioma (images = 2,669). All HCC cases were histopathologically confirmed through surgical resection or biopsy, whereas hemangioma cases were diagnosed radiologically based on characteristic imaging features. Image labels were reviewed and verified by board-certified radiologists, with pathological confirmation for HCC cases.

The full dataset, including class labels and metadata, is publicly available in an external data repository under a CC-BY license. The images are organized in a hierarchical directory structure to facilitate machine learning research. At the top level, the dataset is divided into three subsets: train_set, val_set, and test_set. Within each subset, images are further grouped into class-specific subdirectories (HCC and Hemangioma).

To account for the potential impact of caliper usage during training, the dataset is separated into Caliper and Clean versions. For the Clean version, the specific allocation of images across the training, validation, and test sets is summarized in [Table 1], along with the overall class-wise distribution.

A metadata file in CSV format is provided, with each row corresponding to a single image. The metadata includes the following fields: patient_id, image_filename, class_label, and additional available clinical or acquisition-related information. For HCC cases, the metadata specifically includes the pathologic T stage according to the modified UICC (2000) system, pathologic T stage according to the AJCC (2010) system, and tumor size.

This dataset addresses the limited availability of large-scale, well-annotated ultrasound datasets for liver lesion classification and provides a robust resource for training and validating deep learning models in liver cancer screening and diagnosis.

Patient demographics and tumor characteristics are summarized in [Table 2]. The HCC cohort was older (66.65 ± 11.02 vs 55.50 ± 12.76 years) and predominantly male (81.8% vs 41.3%). Tumor diameter in HCC cases ranged from 2.9, 2.10, 4.5 cm (Median, Q1, Q3). Pathologic staging according to UICC (2000) and AJCC (2010) criteria is provided for HCC cases.

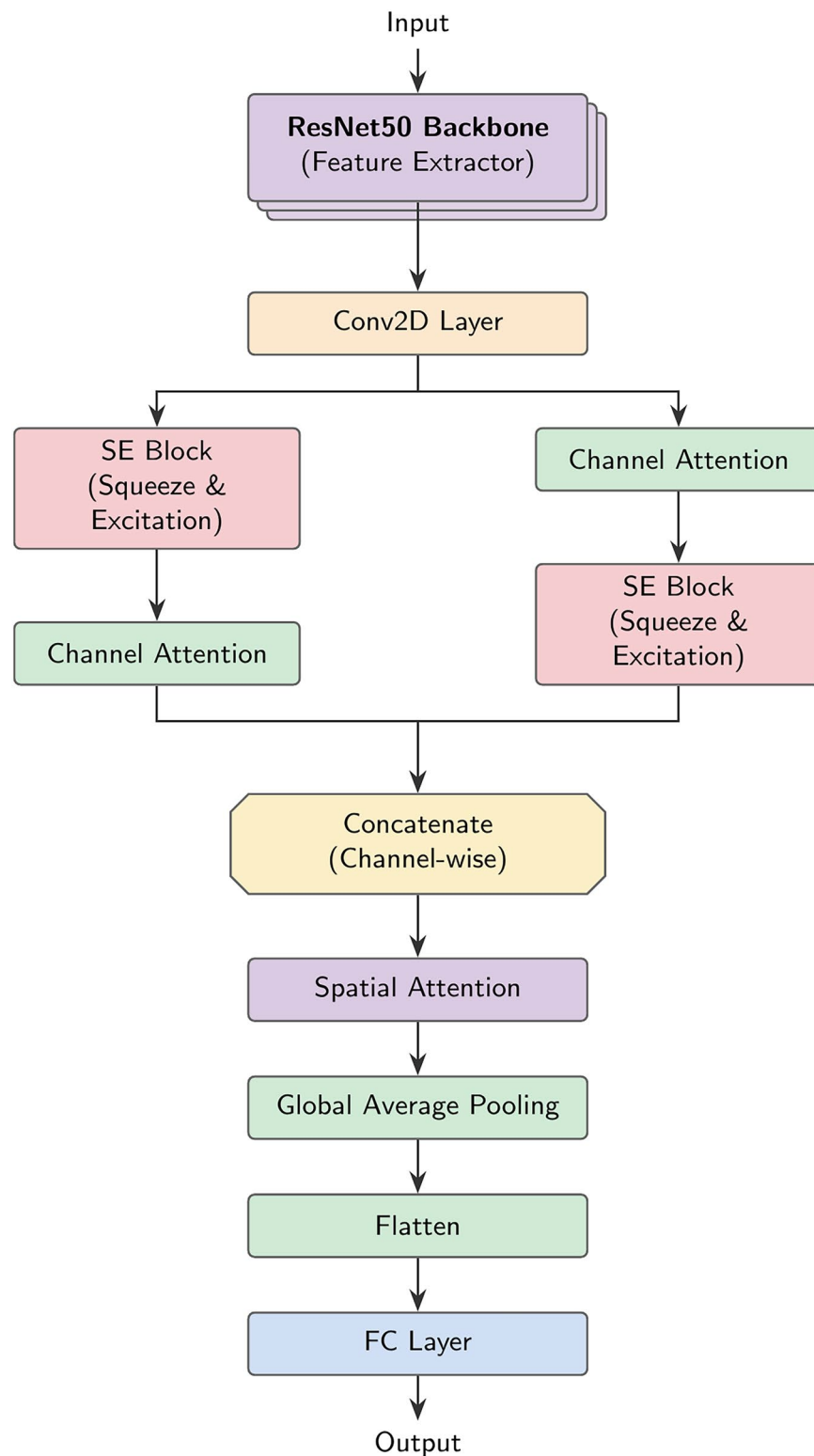


Fig. 2 Parallel attention-enhanced ResNet-50 architecture with dual-branch channel attention and spatial attention module.

Technical Validation

To demonstrate the utility of this dataset, we introduce a novel deep learning model: a parallel attention convolutional neural network (CNN) architecture based on a ResNet-50 backbone. This model incorporates both channel and spatial attention modules through a dual-branch design inspired by squeeze-and-excitation (SE)⁴, bottleneck attention¹², and concurrent spatial-channel excitation blocks¹³. It is designed to enhance lesion localization and fine-grained feature learning, enabling accurate classification into the two target categories. We

Diagnosis	Subgroup	No. of Patients	No. of Images	Train	Validation	Test
Hemangioma	Clean	323	1,267	886	253	128
	Caliper	374	1,402			
	Subtotal	421	2,669			
HCC	Clean	421	1,389	972	277	140
	Caliper	557	1,327			
	Subtotal	600	2,716			
Clean (Trained)		744	2,656	1,858	530	268
Total		1,021	5,385	3,769	1,076	540

Table 1. Dataset Classes.

Characteristic	HCC (patients = 600)	Hemangioma (patients = 421)
Demographics		
Age, years	66.65 ± 11.02 (20–90)	55.50 ± 12.76 (30–90)
Sex, male	491 (81.8%)	174 (41.3%)
Sex, female	109 (18.2%)	247 (58.7%)
Height, cm	165.74 ± 16.39	163.72 ± 8.45
Weight, kg	67.44 ± 11.24 (42–104)	63.85 ± 12.95 (34–118)
Tumor characteristics		
Max diameter, cm	2.90/2.10/4.50(Median/Q1/Q3)	—
UICC stage (2000)		
pT1	58 (9.7%)	—
pT2	120 (20.0%)	—
pT3	353 (58.8%)	—
pT4	69 (11.5%)	—
AJCC stage (2010)		
pT1	164 (27.3%)	—
pT2	393 (65.5%)	—
pT3	22 (3.7%)	—
pT4	21 (3.5%)	—

Table 2. Patient and Tumor Characteristics.

Model	Accuracy	Precision	Recall	F1-Score
Proposed Model	98.88	97.90	100.00	98.94
DenseNet-121	97.39	96.50	98.57	97.53
ResNet-18	96.27	96.43	96.43	96.43
VGG-16	95.90	93.88	98.57	96.17
Swin Transformer (Tiny)	94.40	95.62	93.57	94.58
ResNet-50	92.16	93.43	91.43	92.42
EfficientNet-B0	91.04	91.43	91.43	91.43

Table 3. Performance Comparison (%).

benchmark its performance against several strong baselines, including DenseNet-121, VGG-16, ResNet-18 and ResNet-50, EfficientNet-B0, and the Swin Transformer (Tiny), using standard evaluation metrics such as accuracy, precision, recall, and F1-score².

To evaluate the effectiveness of our approach, we conducted extensive experiments comparing the proposed model against four baselines. [Table 3] summarizes the macro-averaged accuracy, precision, recall, and F1-score on the held-out test set.

The model's performance was compared against other architectures across all metrics. The differentiation between hemangioma and HCC was among the evaluated aspects. Visual inspection of Grad-CAM¹⁹ heatmaps was conducted to observe the model's attention patterns on lesion regions within the images [Figs. 3, 4].

Our proposed model achieved 98.88% accuracy on the test set [Table 3], demonstrating strong discriminative performance on this binary classification task. This result should be interpreted in the context of ultrasound-based liver lesion characterization, where recent systematic reviews report pooled sensitivities and specificities of 80–84% for machine learning-based HCC classification across heterogeneous datasets²⁰. The relatively high performance of our model reflects both the quality of our histopathologically and radiologically confirmed labels and the presence of distinguishable imaging features in B-mode ultrasound for these two

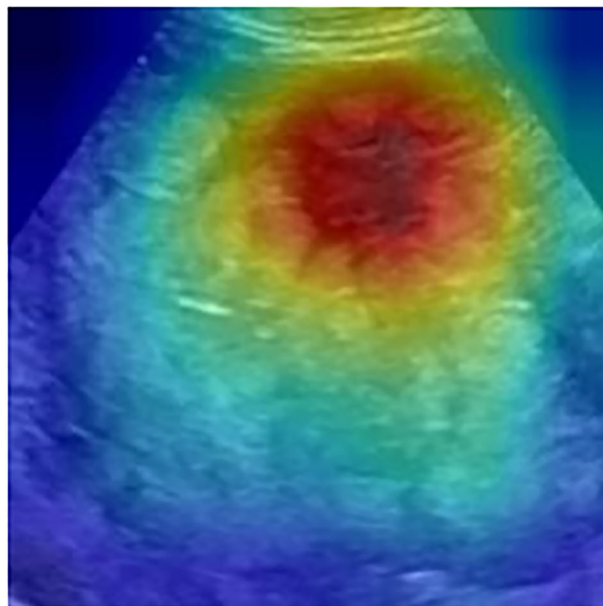


Fig. 3 Grad-CAM visualization showing class activation maps for lesion localization.

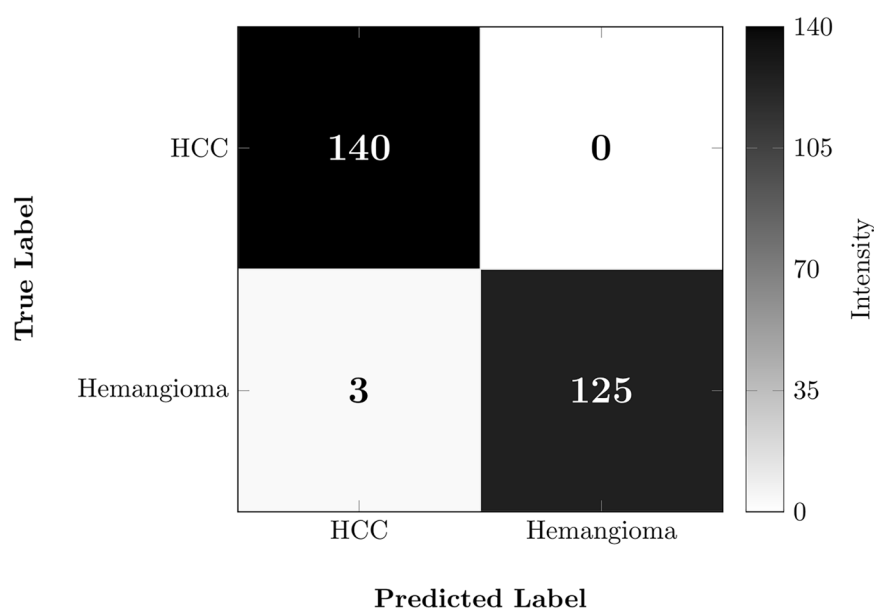


Fig. 4 Confusion matrix on the test set (accuracy: 98.88%). True positives: HCC (n = 140), hemangioma (n = 128).

lesion types. Nevertheless, we acknowledge that binary classification represents only one aspect of the complex diagnostic landscape in liver imaging. The clinical utility and generalizability of our dataset extend beyond the baseline results presented here, as it provides a foundation for more challenging tasks that remain open for future research.

Usage Notes

SMC-LUD is suitable for training and validating deep learning models for liver lesion classification. Application of transfer learning and evaluation of model generalizability across domains are recommended. For clinical translation, attention should be given to domain shifts resulting from variations in ultrasound equipment, operator technique, and patient demographics. Fine-tuning models on site-specific data prior to deployment is advised. As SMC-LUD is designed for binary classification between hepatocellular carcinoma (HCC) and hemangioma, expanding the classification task to include multiple classes may enhance clinical relevance, although this approach introduces additional technical complexity. Incorporating HCC histological grading

according to the Edmondson–Steiner system, which stratifies tumors from well- to poorly differentiated (I–IV) and provides significant prognostic information^{21,22}, represents a promising research direction. Recent evidence indicates that contrast-enhanced ultrasound (CEUS), when combined with radiomics and ensemble learning, can predict these histological grades with area under the curve (AUC) values ranging from 0.84 to 0.93^{23,24}. These findings support the feasibility of ultrasound-based grading. Further extension to additional focal liver lesions, such as focal nodular hyperplasia, metastases, or intrahepatic cholangiocarcinoma, as well as prediction of features including microvascular invasion, could establish more comprehensive benchmarks for algorithm development^{25–27}.

Ethics statement. The Institutional Review Board (IRB) of Samsung Medical Center (SMC) approved this study. The IRB waived the requirement for informed consent because the research was retrospective and used de-identified medical records and ultrasound images previously collected during routine clinical practice. Participants were selected from the SMC database among patients who underwent abdominal ultrasound between January 2015 and December 2024. All data were processed to protect participant privacy: personal identification information was coded, and sensitive DICOM header tags, such as PatientName and PatientID, were removed or replaced using the pydicom library. Only authorized researchers had access to the dataset, which was stored in a secure, access-controlled environment. No identifiable information will be disclosed in any publications.

Data availability

The dataset generated and analyzed during the current study, titled “SMC-LUD,” is publicly available in the Figshare repository at <https://doi.org/10.6084/m9.figshare.31112716>¹⁸.

Due to the strict ethical guidelines and patient privacy protections mandated by the Data Review Board of Samsung Medical Center (SMC) (DRB No. 2025-0111), the data is provided for non-commercial research use only. While the hosting platform may display a default license label, all users must adhere to the non-commercial restriction as specified in the repository’s metadata and this manuscript.

Code availability

All source code for data preprocessing, model training, and evaluation is available on GitHub at: https://github.com/jstak0804/Ultrasound_project. The repository includes scripts to reproduce all experiments in this study and instructions for applying the model to custom ultrasound datasets.

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Author contributions

Conceptualization: J.T., R.-E.K., Z.A., N.O., and S.W.L.; Dataset curation: Y.H.C. and J.K.; Dataset Handling: J.T., R.D.K., and S.S. Model design: J.T., R.-E.K., Z.A., and R.D.K.; Code implementation: J.T., R.-E.K., Z.A., and R.D.K.; Experimentation: J.T., R.-E.K., and Z.A.; Analysis: J.T., R.-E.K., Z.A., and S.W.L.; Manuscript writing: J.T., R.-E.K., Z.A., and R.D.K.; Figure creation: J.T., R.-E.K., Z.A., and R.D.K.; Supervision: N.O. and S.W.L.; Manuscript review and editing: R.D.K., N.O. and S.W.L. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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